Randomized Algorithms (2026 Spring)
Quiz 2 – Detailed Solutions

This solution set corresponds to the uploaded Quiz 2 and the related lecture notes.

1. **Suppose that S is a set of $k > 0$ integers such that $k = \lceil n/\log n \rceil$. Prove or disprove that sorting S by an optimal comparison-based algorithm can be done in $o(n)$ time.**

   Solution. The statement is *false*.

   An optimal comparison-based sorting algorithm takes

$$\Theta(k \log k)$$

   time in the worst case. Since

$$k = \left\lceil \frac{n}{\log n} \right\rceil,$$

   we have asymptotically

$$k = \Theta\left(\frac{n}{\log n}\right).$$

   Therefore,

$$k \log k = \Theta\left(\frac{n}{\log n} \cdot \log\left(\frac{n}{\log n}\right)\right).$$

   Now,

$$\log\left(\frac{n}{\log n}\right) = \log n - \log\log n,$$

   so

$$k \log k = \Theta\left(\frac{n}{\log n}(\log n - \log\log n)\right) = \Theta\left(n - \frac{n \log\log n}{\log n}\right).$$

   To see why this last quantity is still $\Theta(n)$, factor out n:

$$n - \frac{n \log\log n}{\log n} = n\left(1 - \frac{\log\log n}{\log n}\right).$$

   To justify

$$\frac{\log\log n}{\log n} \to 0 \qquad \text{as } n \to \infty,$$

   let

$$x = \log n.$$

   Then $x \to \infty$ as $n \to \infty$, and the ratio becomes

$$\frac{\log\log n}{\log n} = \frac{\log x}{x}.$$

Now $\log x$ grows much more slowly than x. For instance, by L'Hôpital's rule,

$$\lim_{x\to\infty} \frac{\log x}{x} = \lim_{x\to\infty} \frac{1/x}{1} = \lim_{x\to\infty} \frac{1}{x} = 0.$$

Hence

$$\frac{\log\log n}{\log n} \to 0.$$

Therefore,

$$1 - \frac{\log\log n}{\log n} \to 1.$$

So for all sufficiently large n, this factor is bounded above and below by positive constants; for example, there exists n_0 such that for all $n \geq n_0$,

$$\frac{1}{2} \leq 1 - \frac{\log\log n}{\log n} \leq 1.$$

Multiplying by n, we get

$$\frac{n}{2} \leq n\left(1 - \frac{\log\log n}{\log n}\right) \leq n \qquad \text{for all sufficiently large } n.$$

Therefore,

$$n - \frac{n\log\log n}{\log n} = \Theta(n),$$

and so

$$k\log k = \Theta(n).$$

Hence sorting k elements takes

$$\Theta(k\log k) = \Theta(n),$$

which is *not* $o(n)$.

Conclusion.

$$\boxed{\text{Disprove: sorting } k = \left\lceil \frac{n}{\log n} \right\rceil \text{ elements requires } \Theta(n) \text{ time, not } o(n).}$$

2. **Suppose that we roll a standard fair die 120 times. Let X be the sum of the numbers that appear over the 120 rolls. Use Chebyshev's inequality to bound $\Pr[|X - 420| \geq 100]$.**

Solution. Let

$$X = X_1 + X_2 + \cdots + X_{120},$$

where each X_i is the outcome of the i-th die roll.

Since the die is fair,

$$\mathbb{E}[X_i] = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = \frac{7}{2}.$$

Thus

$$\mathbb{E}[X] = 120 \cdot \frac{7}{2} = 420.$$

Next, compute the variance of one die roll. We have

$$\mathbb{E}[X_i^2] = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2}{6} = \frac{91}{6}.$$

Hence

$$\mathrm{Var}(X_i) = \mathbb{E}[X_i^2] - (\mathbb{E}[X_i])^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}.$$

Because the 120 rolls are independent,

$$\mathrm{Var}(X) = 120 \cdot \frac{35}{12} = 10 \cdot 35 = 350.$$

By Chebyshev's inequality,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq \frac{\mathrm{Var}(X)}{t^2}.$$

Taking $t = 100$, we get

$$\mathbb{P}(|X - 420| \geq 100) \leq \frac{350}{100^2} = \frac{350}{10000} = \frac{7}{200}.$$

Conclusion.

$$\boxed{\mathbb{P}(|X - 420| \geq 100) \leq \frac{7}{200}.}$$

3. **A simple model of the stock market suggests that, each day, a stock with price q will increase by a factor $r > 1$ to qr with probability p and will fall to q/r with probability $1 - p$. Assuming we start with a stock with price 1, find a formula for the expected value and the variance of the price of the stock after d days.**

Solution. Let Y_i denote the multiplicative factor on day i:

$$Y_i = \begin{cases} r, & \text{with probability } p, \\ 1/r, & \text{with probability } 1 - p. \end{cases}$$

Assume the daily changes are independent. If P_d denotes the stock price after d days, then

$$P_d = \prod_{i=1}^{d} Y_i.$$

Expected value. By independence,

$$\mathbb{E}[P_d] = \mathbb{E}\left[\prod_{i=1}^{d} Y_i\right] = \prod_{i=1}^{d} \mathbb{E}[Y_i] = (\mathbb{E}[Y_1])^d.$$

Now

$$\mathbb{E}[Y_1] = pr + (1 - p)\frac{1}{r}.$$

Therefore,

$$\boxed{\mathbb{E}[P_d] = \left(pr + \frac{1-p}{r}\right)^d.}$$

Second moment. Similarly,

$$\mathbb{E}[P_d^2] = \mathbb{E}\left[\prod_{i=1}^{d} Y_i^2\right] = \prod_{i=1}^{d} \mathbb{E}[Y_i^2] = \left(\mathbb{E}[Y_1^2]\right)^d.$$

Since

$$\mathbb{E}[Y_1^2] = pr^2 + (1-p)\frac{1}{r^2},$$

we get

$$\mathbb{E}[P_d^2] = \left(pr^2 + \frac{1-p}{r^2}\right)^d.$$

Variance. Using $\mathrm{Var}(P_d) = \mathbb{E}[P_d^2] - (\mathbb{E}[P_d])^2$, we obtain

$$\boxed{\mathrm{Var}(P_d) = \left(pr^2 + \frac{1-p}{r^2}\right)^d - \left(pr + \frac{1-p}{r}\right)^{2d}.}$$

4. **Let S be a set of $n = 2k+1$ integers for some positive integer k. If we pick three numbers x, y, z from S uniformly at random (with replacement), what is the probability that the median of $\{x, y, z\}$ is the median of S?**

Solution. Let m be the true median of S. Since $|S| = 2k+1$, there are exactly:

- k elements smaller than m,
- 1 element equal to m,
- k elements larger than m.

We count the favorable ordered triples (x, y, z).

The sample median equals m in exactly the following cases.

Case 1: Exactly one of x, y, z equals m. Then one of the other two values must be smaller than m, and the other must be larger than m.

The number of such triples is:

$$3 \cdot 2 \cdot k \cdot k = 6k^2.$$

Indeed:

- choose the position of m: 3 ways;
- choose which of the remaining two positions gets the smaller-side element: 2 ways;
- choose the smaller-side element: k ways;
- choose the larger-side element: k ways.

Case 2: Exactly two of x, y, z equal m. Then the third draw can be any non-median element, of which there are $2k$. Thus the number of such triples is

$$3 \cdot 2k = 6k,$$

because we choose the position of the non-median element in 3 ways.

Case 3: All three equal m. This contributes exactly 1 favorable triple.

Hence the total number of favorable ordered triples is

$$6k^2 + 6k + 1.$$

Since sampling is with replacement, the total number of ordered triples is

$$(2k+1)^3.$$

Therefore,

$$\boxed{\Pr[\text{the median of } \{x, y, z\} \text{ is the median of } S] = \frac{6k^2 + 6k + 1}{(2k+1)^3}.}$$

In terms of n, where $n = 2k + 1$, this can be rewritten as

$$6k^2 + 6k + 1 = \frac{3n^2 - 1}{2},$$

so equivalently,

$$\boxed{\Pr[\text{success}] = \frac{3n^2 - 1}{2n^3}.}$$

5. **Suppose that S is a set of $n = 2k + 1$ distinct numbers for some integer $k > 0$. Let z be a number picked uniformly at random from S and define $S_1 := \{x \in S : x < z\}$, $S_2 := \{x \in S : x > z\}$. Please show that $\mathbb{E}[\max\{|S_1|, |S_2|\}] \leq \frac{3}{4}n$.**

Solution. Because all elements are distinct, each element of S has a unique rank in sorted order. Let the rank of z be j, where

$$j \in \{1, 2, \ldots, 2k + 1\}.$$

Since z is chosen uniformly from S, every rank is equally likely:

$$\Pr[\text{rank}(z) = j] = \frac{1}{2k+1}.$$

If z has rank j, then:

$$|S_1| = j - 1, \qquad |S_2| = 2k + 1 - j.$$

Therefore,

$$\max\{|S_1|, |S_2|\} = \max\{j - 1,\ 2k + 1 - j\}.$$

So

$$\mathbb{E}[\max\{|S_1|, |S_2|\}] = \frac{1}{2k+1} \sum_{j=1}^{2k+1} \max\{j - 1, 2k + 1 - j\}.$$

Now list these values explicitly as j runs from 1 to $2k+1$:

$$2k,\ 2k-1,\ \ldots,\ k+1,\ k,\ k+1,\ \ldots,\ 2k-1,\ 2k.$$

Hence the sum equals

$$2\sum_{t=k+1}^{2k} t + k.$$

We compute

$$\sum_{t=k+1}^{2k} t = \frac{(k+1+2k)k}{2} = \frac{(3k+1)k}{2}.$$

Therefore,

$$2\sum_{t=k+1}^{2k} t + k = (3k+1)k + k = 3k^2 + 2k.$$

Thus

$$\mathbb{E}[\max\{|S_1|, |S_2|\}] = \frac{3k^2+2k}{2k+1}.$$

Now compare this with $\frac{3}{4}n = \frac{3}{4}(2k+1)$. We need to show

$$\frac{3k^2+2k}{2k+1} \leq \frac{3}{4}(2k+1).$$

Multiplying both sides by $4(2k+1) > 0$ gives

$$4(3k^2+2k) \leq 3(2k+1)^2.$$

Expanding both sides,

$$12k^2 + 8k \leq 12k^2 + 12k + 3,$$

which is clearly true. Hence

$$\boxed{\mathbb{E}[\max\{|S_1|, |S_2|\}] \leq \frac{3}{4}n.}$$

In fact, we have the exact value

$$\mathbb{E}[\max\{|S_1|, |S_2|\}] = \frac{3k^2+2k}{2k+1} = \frac{3n}{4} - \frac{1}{2} - \frac{1}{4n},$$

which is strictly smaller than $\frac{3}{4}n$.